Dataset	$\#F_{total}$	$min(\|Var(F)\|)$	$max(\|Var(F)\|)$	$min(\|F\|)$	$max(\|F\|)$
Global	4550	0	8286433	0	7689680
↪ Lagniez [?]	1948	5	229100	10	399794
↪↪ Bayesian Network	1085	32	229100	38	399794
↪↪ BMC	18	762	63624	2469	368352
↪↪ Circuit	68	26	8704	66	83902
↪↪ Configuration	35	1400	2038	1698	11342
↪↪ Handmade	68	61	3176	254	174160
↪↪ Planning	557	5	24816	10	148891
↪↪ QIF	7	251	4473	452	14011
↪↪ Random	104	75	250	150	1065
↪↪ Scheduling	6	19500	22500	103887	123329
↪ Soos [?]	1823	2	8286433	1	7689680
↪ Plazar [?]	501	14	486193	31	2598178
↪↪ Blasted Real	210	14	10329	35	33008
↪↪ Feature Models	133	45	62482	104	343944
↪↪ V15	30	17	25615	43	57946
↪↪ V3	30	17	25528	31	57586
↪↪ V7	30	17	25546	35	57662
↪↪ unsorted	68	67	486193	66	2598178
↪ Sundermann [?]	278	0	31012	0	350120

Table 1: Dataset summary. The first column indicates the dataset, and the $\#F_{total}$ column indicates how many satisfiable formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

Dataset	$\#F_{total}$	$\#D4$	$\#\neg D4$	$\#CapKC \land \neg D4$	$\#DivKC \land \neg CapKC \land \neg D4$
Global	4550	185	670	80	55
↪ Lagniez [?]	1948	53	213	40	23
↪↪ Bayesian Network	1085	16	70	37	22
↪↪ BMC	18	12	2	0	0
↪↪ Circuit	68	1	8	0	0
↪↪ Configuration	35	1	0	0	0
↪↪ Handmade	68	0	32	1	0
↪↪ Planning	557	23	92	1	1
↪↪ QIF	7	0	1	0	0
↪↪ Random	104	0	2	1	0
↪↪ Scheduling	6	0	6	0	0
↪ Soos [?]	1823	100	393	40	28
↪ Plazar [?]	501	31	61	0	4
↪↪ Blasted Real	210	6	30	0	3
↪↪ Feature Models	133	1	5	0	0
↪↪ V15	30	1	5	0	0
↪↪ V3	30	1	5	0	0
↪↪ V7	30	1	5	0	1
↪↪ unsorted	68	21	11	0	0
↪ Sundermann [?]	278	1	3	0	0

Table 2: Experimental results regarding the scalability of DivKC and CapKC. Column $\#F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by DivKC or CapKC. Column $\#\neg D4$ shows the number of formulae not compiled by D4. Column $\#CK \land \neg D4$ shows the number of formulae that are compiled by CapKC but not by D4. The last column indicates the number of formulae that were only compiled by DivKC, but not by D4 or CapKC.

Dataset	$\#F_{total}$	$\#\neg D4$	$\#\neg D4 \wedge CapKC$	$\#\neg D4 \wedge DivKC$	$\#CapKC \wedge \neg DivKC \wedge \neg D4$
Global	4550	670	80	114	21
↪ Lagniez [?]	1948	213	40	52	11
↪↪ Bayesian Network	1085	70	37	51	8
↪↪ BMC	18	2	0	0	0
↪↪ Circuit	68	8	0	0	0
↪↪ Configuration	35	0	0	0	0
↪↪ Handmade	68	32	1	0	1
↪↪ Planning	557	92	1	1	1
↪↪ QIF	7	1	0	0	0
↪↪ Random	104	2	1	0	1
↪↪ Scheduling	6	6	0	0	0
↪ Soos [?]	1823	393	40	58	10
↪ Plazar [?]	501	61	0	4	0
↪↪ Blasted Real	210	30	0	3	0
↪↪ Feature Models	133	5	0	0	0
↪↪ V15	30	5	0	0	0
↪↪ V3	30	5	0	0	0
↪↪ V7	30	5	0	1	0
↪↪ unsorted	68	11	0	0	0
↪ Sundermann [?]	278	3	0	0	0

Table 3: Experimental results regarding the scalability of DivKC and CapKC.

Dataset	#F	$Y_l \leq \|R_F\|$	$Y_h \geq \|R_F\|$	Coverage	$\|R_{G_P}\| \leq \|R_F\| \leq \|R_{G_U}\|$
Global	3670	0.969	0.857	0.826	1.000
↪ Lagniez [?]	1689	0.969	0.895	0.864	1.000
↪↪ Bayesian Network	1009	0.985	0.908	0.893	1.000
↪↪ BMC	4	1.000	1.000	1.000	1.000
↪↪ Circuit	59	1.000	0.814	0.814	1.000
↪↪ Configuration	33	0.788	0.697	0.485	1.000
↪↪ Handmade	35	1.000	1.000	1.000	1.000
↪↪ Planning	441	0.932	0.862	0.794	1.000
↪↪ QIF	6	1.000	0.833	0.833	1.000
↪↪ Random	102	0.990	0.990	0.980	1.000
↪↪ Scheduling	0				
↪ Soos [?]	1307	0.969	0.906	0.875	1.000
↪ Plazar [?]	403	0.968	0.789	0.757	1.000
↪↪ Blasted Real	176	1.000	0.938	0.938	1.000
↪↪ Feature Models	124	0.968	0.460	0.427	1.000
↪↪ V15	23	1.000	1.000	1.000	1.000
↪↪ V3	23	1.000	1.000	1.000	1.000
↪↪ V7	23	1.000	1.000	1.000	1.000
↪↪ unsorted	34	0.735	0.794	0.529	1.000
↪ Sundermann [?]	271	0.967	0.487	0.454	1.000

Table 4: Experimental results for DivKC-AMC. Column #F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	#F	$Y_l \leq \lvert R_F \rvert$	$Y_h \geq \lvert R_F \rvert$	Coverage	$\lvert R_F \rvert \leq \lvert R_{G_U} \rvert$
Global	323	0.994	0.988	0.981	1.000
↪ Lagniez [?]	168	0.994	0.988	0.982	1.000
↪↪ Bayesian Network	138	0.993	0.993	0.986	1.000
↪↪ BMC	0				
↪↪ Circuit	2	1.000	1.000	1.000	1.000
↪↪ Configuration	8	1.000	0.875	0.875	1.000
↪↪ Handmade	1	1.000	1.000	1.000	1.000
↪↪ Planning	17	1.000	1.000	1.000	1.000
↪↪ QIF	1	1.000	1.000	1.000	1.000
↪↪ Random	1	1.000	1.000	1.000	1.000
↪↪ Scheduling	0				
↪ Soos [?]	149	1.000	0.987	0.987	1.000
↪ Plazar [?]	4	1.000	1.000	1.000	1.000
↪↪ Blasted Real	3	1.000	1.000	1.000	1.000
↪↪ Feature Models	0				
↪↪ V15	0				
↪↪ V3	0				
↪↪ V7	0				
↪↪ unsorted	1	1.000	1.000	1.000	1.000
↪ Sundermann [?]	2	0.500	1.000	0.500	1.000

Table 5: Experimental results for CapKC. Column #F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	#F	$Y_l \geq \lvert R_{G_P} \rvert$	$Y_h \leq \lvert R_{G_U} \rvert$	Both	$median(r_c)$	$max(r_c)$
Global	3669	0.807	0.836	0.721	4.17e-9	0.54
↪ Lagniez [?]	1689	0.844	0.869	0.760	1.8e-9	0.54
↪↪ Bayesian Network	1009	0.935	0.899	0.850	8.16e-7	0.0963
↪↪ BMC	4	1.000	1.000	1.000	0.0	2.34e-5
↪↪ Circuit	59	0.932	0.814	0.797	5.0e-30	4.27e-7
↪↪ Configuration	33	0.394	0.485	0.303	4.54e-6	0.54
↪↪ Handmade	35	1.000	1.000	1.000	0.0	0.0
↪↪ Planning	441	0.834	0.798	0.735	7.59e-40	0.0181
↪↪ QIF	6	0.667	0.833	0.500	1.12e-56	1.89e-5
↪↪ Random	102	0.029	0.990	0.029	1.84e-68	1.92e-23
↪↪ Scheduling	0					
↪ Soos [?]	1307	0.920	0.893	0.834	1.75e-7	0.0962
↪ Plazar [?]	403	0.697	0.769	0.610	1.64e-13	0.0153
↪↪ Blasted Real	176	0.932	0.938	0.881	2.99e-13	0.0153
↪↪ Feature Models	124	0.202	0.427	0.048	3.8e-13	0.00975
↪↪ V15	23	1.000	1.000	1.000	2.51e-17	0.00603
↪↪ V3	23	1.000	1.000	1.000	2.92e-17	0.00401
↪↪ V7	23	0.957	1.000	0.957	2.35e-17	0.00406
↪↪ unsorted	34	0.706	0.676	0.500	7.57e-91	2.39e-9
↪ Sundermann [?]	270	0.196	0.448	0.093	1.69e-13	0.281

Table 6: Experimental results comparing the bounds obtained with DivKC-AMC and with Lemma **??**. Column #F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$ and $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$. The last two columns represent the observed median and maximum values of the ratio $r_c = \frac{min(Y_h, |R_{G_U}|) - max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$, which was calculated exclusively if $Y_l \leq |R_F| \leq Y_h$. The number of formulae on which the last two columns are computed can easily be obtained by multiplying the #F column with the 'Coverage' column in Table 4.

Dataset	#F	$Y_h \leq \lvert R_{G_U} \rvert$	$median(r_c)$	$max(r_c)$
Global	334	0.994	0.00812	0.0257
↪ Lagniez [?]	175	0.994	0.00753	0.0257
↪↪ Bayesian Network	143	1.000	0.00915	0.0257
↪↪ BMC	0			
↪↪ Circuit	2	1.000	0.00757	0.0103
↪↪ Configuration	8	0.875	0.00996	0.0256
↪↪ Handmade	2	1.000	0.0135	0.0231
↪↪ Planning	18	1.000	0.00164	0.0257
↪↪ QIF	1	1.000	0.0257	0.0257
↪↪ Random	1	1.000	0.00422	0.00422
↪↪ Scheduling	0			
↪ Soos [?]	153	0.993	0.00861	0.0257
↪ Plazar [?]	4	1.000	0.00684	0.0257
↪↪ Blasted Real	3	1.000	0.00339	0.0257
↪↪ Feature Models	0			
↪↪ V15	0			
↪↪ V3	0			
↪↪ V7	0			
↪↪ unsorted	1	1.000	0.0103	0.0103
↪ Sundermann [?]	2	1.000	0.00981	0.0155

Table 7: Experimental results comparing the bounds obtained with CapKC. Column #F indicates with how many formulae the statistics have been computed. The last two columns represent the observed median and maximum values of the ratio $r_c = \dfrac{Y_h - Y_l}{|R_{G_U}|}$, which was calculated exclusively if $Y_l \leq |R_F| \leq Y_h$. The number of formulae on which the last two columns are computed can easily be obtained by multiplying the #F column with the 'Coverage' column in Table 5.

Dataset	#F	$l \leq Y_{ApproxMC} \leq h$	$l \leq Y_{DivKC\text{-}AMC} \leq h$	$l \leq Y_{CapKC\text{-}AMC} \leq h$
Global	2713	0.973	0.871	0.981
↪ Lagniez [?]	1329	1.000	0.889	0.980
↪↪ Bayesian Network	922	1.000	0.851	0.975
↪↪ BMC	3	1.000	1.000	1.000
↪↪ Circuit	44	1.000	0.864	1.000
↪↪ Configuration	16	1.000	0.875	1.000
↪↪ Handmade	12	1.000	1.000	1.000
↪↪ Planning	319	1.000	0.994	0.991
↪↪ QIF	5	1.000	0.800	1.000
↪↪ Random	8	1.000	1.000	1.000
↪↪ Scheduling	0			
↪ Soos [?]	1177	0.964	0.847	0.979
↪ Plazar [?]	185	0.838	0.892	0.995
↪↪ Blasted Real	149	1.000	0.886	0.993
↪↪ Feature Models	6	1.000	1.000	1.000
↪↪ V15	0			
↪↪ V3	0			
↪↪ V7	0			
↪↪ unsorted	30	0.000	0.900	1.000
↪ Sundermann [?]	22	1.000	0.955	1.000

Table 8: CapKC.

Dataset	#F	#DivKC-AMC	$log_{10}(min)$	$mean$	$median$	$log_{10}(max)$
Global	3188	2129	-3.7	160.7	1.9	5.1
↪ Lagniez [?]	1599	1160	-3.0	239.1	2.1	4.9
↪↪ Bayesian Network	1051	708	-2.7	7.9	1.9	3.8
↪↪ BMC	4	0	-1.5	0.3	0.3	-0.3
↪↪ Circuit	52	28	-2.9	1.7	1.4	1.2
↪↪ Configuration	16	16	0.5	5692.8	50.7	4.9
↪↪ Handmade	15	14	-0.2	220.0	11.2	3.4
↪↪ Planning	352	288	-3.0	749.4	4.7	4.6
↪↪ QIF	7	5	-0.6	2.4	3.0	0.6
↪↪ Random	102	101	-0.4	155.1	101.9	2.7
↪↪ Scheduling	0					
↪ Soos [?]	1349	855	-3.6	6.2	1.8	3.6
↪ Plazar [?]	218	97	-3.7	1.3	0.9	1.2
↪↪ Blasted Real	177	85	-3.3	1.2	0.9	0.7
↪↪ Feature Models	6	4	-0.3	4.7	3.1	1.2
↪↪ V15	0					
↪↪ V3	0					
↪↪ V7	0					
↪↪ unsorted	35	8	-3.7	1.1	0.3	1.0
↪ Sundermann [?]	22	17	-1.0	5507.7	12.6	5.1

Table 9: DivKC vs Approxmc7.

Dataset	#F	#CapKC-AMC	$log_{10}(min)$	$mean$	$median$	$log_{10}(max)$
Global	2514	1554	-2.3	179.1	1.8	5.0
↪ Lagniez [?]	1205	773	-2.3	287.9	2.0	5.0
↪↪ Bayesian Network	865	507	-2.2	18.6	1.7	4.2
↪↪ BMC	0					
↪↪ Circuit	40	25	-1.4	1.8	1.7	0.8
↪↪ Configuration	16	16	0.5	6890.2	301.7	5.0
↪↪ Handmade	4	0	-1.1	0.2	0.1	-0.4
↪↪ Planning	274	223	-1.4	804.9	6.0	4.4
↪↪ QIF	5	2	-1.0	1.4	0.1	0.6
↪↪ Random	1	0	-2.3	0.0	0.0	-2.3
↪↪ Scheduling	0					
↪ Soos [?]	1111	651	-2.2	3.3	1.6	2.8
↪ Plazar [?]	176	111	-1.2	2.1	1.5	1.1
↪↪ Blasted Real	139	96	-1.2	2.2	1.7	1.1
↪↪ Feature Models	6	5	-0.5	4.6	3.7	1.1
↪↪ V15	0					
↪↪ V3	0					
↪↪ V7	0					
↪↪ unsorted	31	10	-1.1	1.0	0.3	0.8
↪ Sundermann [?]	22	19	-0.9	4517.2	11.1	5.0

Table 10: CapKC vs Approxmc7.

Dataset	#F	#CapKC-AMC	$log_{10}(min)$	$mean$	$median$	$log_{10}(max)$
Global	421	182	-2.2	250.3	0.6	5.0
↪ Lagniez [?]	207	92	-2.2	506.3	0.6	5.0
↪↪ Bayesian Network	166	58	-2.2	1.2	0.4	0.9
↪↪ BMC	0					
↪↪ Circuit	2	2	0.4	2.5	2.5	0.4
↪↪ Configuration	2	2	3.6	49836.7	49836.7	5.0
↪↪ Handmade	0					
↪↪ Planning	35	29	-1.0	140.5	3.3	3.5
↪↪ QIF	2	1	-0.8	1.9	1.9	0.6
↪↪ Random	0					
↪↪ Scheduling	0					
↪ Soos [?]	194	74	-2.2	2.7	0.5	2.2
↪ Plazar [?]	16	12	-0.9	2.8	2.1	0.7
↪↪ Blasted Real	15	11	-0.9	2.7	1.8	0.7
↪↪ Feature Models	0					
↪↪ V15	0					
↪↪ V3	0					
↪↪ V7	0					
↪↪ unsorted	1	1	0.7	5.5	5.5	0.7
↪ Sundermann [?]	4	4	0.0	5.6	6.6	0.9

Table 11: CapKC vs Approxmc7.